

Ajit Thapa

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DevOps Engineer

Professional Summary

DevOps Engineer with over 4 years of experience specializing in cloud architecture, infrastructure automation, and CI/CD pipeline development. Skilled in deploying and maintaining secure, scalable, and reliable systems in cloud environments, with a focus on Azure and AWS. Proficient in containerization and orchestration technologies, including Docker and Kubernetes, and experienced in infrastructure as code (IaC) with tools like Terraform and Ansible. Demonstrated success in automating deployments, optimizing CI/CD workflows, and supporting cross-functional teams to improve system reliability and efficiency. Strong background in Linux system administration, network troubleshooting, and application monitoring using tools like Prometheus, Grafana, and Splunk.

Core Competencies:

- Cloud Platforms: Extensive experience with Azure and AWS, including managing SaaS, PaaS, and IaaS solutions and migrating on-premises infrastructure to the cloud.
- CI/CD Pipelines: Proficient in setting up and maintaining CI/CD pipelines using Azure DevOps, Jenkins, and GitHub to streamline code deployment and improve delivery efficiency.
- Containerization and Orchestration: Skilled in Docker, Docker Swarm, and Kubernetes for container management, clustering, and microservice orchestration.
- Infrastructure Automation: Expert in using Terraform, Ansible, and Chef to automate provisioning and configuration of cloud and on-premises infrastructure.
- Monitoring and Logging: Strong experience with Prometheus, Grafana, and Splunk for system and application performance monitoring, creating custom alerts, and maintaining high availability.
- Scripting and Automation: Developed scripts in Shell, PowerShell, and Python to automate tasks, improve operational workflows, and manage configurations.
- Linux System Administration: Solid foundation in Linux (RHEL, CentOS, Ubuntu) system management, including network configurations, security, and troubleshooting.

Education Details

Masters in Information Technology

University of the Cumberland, Williamsburg KY

Graduated: May 2024, 3.89 GPA

Bachelor's of Science : Finance

McNeese State University, Lake Charles LA

Graduated: Dec 2021

Technical Skills:

CI/CD	Azure DevOps, Jenkins, VSTS, GitHub/Bitbucket, Maven, ANT, Confluence, JIRA,
Cloud Platforms	Amazon Web services, Azure
Microservices	Docker, Kubernetes, AWS ECS
Logging and Alerting	Splunk
Distributed Messaging Platform	KAFKA

Infrastructure Automation	Ansible, Terraform, Chef
Test Automation	Selenium
Application Servers	Apache Tomcat,
Web Servers	Nginx
Operating Systems	Windows, Linux, CentOS, Ubuntu,
Databases	Mongo DB, MySQL
Scripting	Python, Shell, PowerShell, JavaScript, YAML

Professional Experience:

DevOps Engineer/ March 2022 – Present

Flight Safety International, Columbus OH (Remote)

Responsibilities:

- Control and maintain all the Docker swarm environments for smooth and efficient deployments of apps and services.
- CI/CD Pipeline Maintenance: Supporting the creation and maintenance of Continuous Integration/Continuous Deployment (CI/CD) pipelines, ensuring smooth delivery of code updates.
- Monitoring and Alerts: Setting up and maintaining monitoring systems (e.g., Prometheus, Pager duty) and responding to system alerts.
- Version Control: Managing code repositories (e.g., Git) and ensuring best practices for version control, including branching and merging strategies.
- Environment Setup: Assisting in the creation, management, and optimization of development, testing, and production environments, mostly using Docker and sometimes Kubernetes.
- Collaboration with Developers: Working closely with development teams to ensure smooth code integration and deployment, addressing issues that may arise. (On-call support for developers)
- Scripting and Automation: Inheriting scripts (PS, YAML, etc.) from the team and use it to automate repetitive tasks and improve operational efficiency.
- System Administration: Supporting system administration tasks such as User-Access (Mostly In Azure DevOps), server maintenance, and troubleshooting.
- Security Best Practices: Helping implement security policies, such as firewall rules, access controls, and vulnerability scanning.
- Production Support: On- call prod support for all non-prod and Go-Live deployments.
- Documentation: Writing and maintaining technical documentation for processes, configurations, and tools used.
- Cloud Services Management: Assisting my team in managing cloud infrastructure on platforms like Azure.
- Testing: Work with QA testers and help them run multiple tests in Selenium grid using Azure DevOps
- Planned IT services recovery and resumption and established procedures for conducting post-resumption reviews. Facilitated the documentation of disaster recovery and service contingency plans. Ensured that all IT systems were complying and contained comprehensive plans for the recovery of assets and services.
- Used tracking tool Azure DevOps Boards and triaged the issues and maintained bugs. Maintained Technical Stories for tracking and updating project defects insisted tasks ensuring the successful completion of tasks in sprint.
- Troubleshooting the network issues involving TCP/IP, LAN/WAN networking, DNS, DHCP, SMTP, SNMP, netstat and monitored the servers and Linux scripts regularly and performed troubleshooting.
- Creating a backup for all the instances in Kafka Environment.
- Experience managing Kafka clusters as docker containers on Linux environment.
- Designed and implemented by configuring Topics in Kafka cluster in all environments.
- Review/Merge/Remove Git Pull Requests to bit bucket. Hands on with Git / GitHub for code version control using Branching, Tagging, Release Activities.
- Worked with Docker on multiple cloud providers, from helping developers build and containerize their application (CI/CD) to deploying either on public or private cloud.
- Played a major role in building Docker images, creating build and deployment configurations to deploy applications on On-prem and on the cloud.
- Built Grafana dashboards for viewing network traffic and application logs by using Prometheus data sources.
- Created alerts and monitoring dashboards using Prometheus and Grafana for all microservices deployed on Azure and on-prem.
- Involved in supporting cloud instances running Linux and Windows on Azure, experience with Security Groups and Virtual Private Cloud in AWS.
- Worked on Docker-Compose, Docker-Machine create Docker containers for testing applications in the QA environment and automated the deployments, scaling and management of containerized applications across clusters of hosts.
- Wrote customized Prometheus queries to retrieve data. Made changes to Prometheus collectors and filters ensuring all relevant data are passed into Grafana.

Environment: Azure DevOps, Amazon Web Services (AWS), Docker Swarm, Maven, Ansible, Tomcat, Shell, GIT, Linux.

DevOps Engineer Intern May 2021- Aug 2021

Texas Instruments, Dallas TX

Responsibilities:

- Working as cloud Engineer, involved in configuration for Web apps/Function apps, Application gateway, App Insights, Active directory, Azure Key Vault, Encryption and Security on Azure.
- Collaborated in a fast-paced DevOps team to streamline CI/CD pipelines using Azure DevOps Services and VSTS.
- Used Ansible as Configuration management tool, to automate repetitive tasks, quickly deploys critical applications, and proactively manages change.
- Managed source control and versioning through Git and GitHub, ensuring seamless integration and version tracking across multiple environments.
- Installed and developed different POC's for different application/infrastructure teams both in Apache Kafka and Confluent open source for multiple clients.
- Experience on Azure Site Recovery, Operations Management Suite, Power shell Scripts and ARM templates.
- Automated infrastructure provisioning with Terraform, enhancing scalability and reducing deployment times.
- Developed scripts in PowerShell and Bash for task automation and configuration management on Linux environments.
- Created and maintained YAML configurations for pipeline orchestration and improved pipeline efficiency.
- Worked on creating Docker containers and Docker consoles for managing the application lifecycle. Worked on various Docker components like Docker Engine, Hub, Machine, Compose and Docker Registry.
- Coordinated with cross-functional teams using JIRA to track project progress and manage tasks
- Created Docker images from scratch and customized and modified base images from existing environment configurations and maintaining the image repository for development teams.
- Coordinated and managed resources responsible for the development, validation and maintenance of Security, Disaster Recovery, and Capability plans. Minimized the impact of major system outages to critical patient care and business operations.
- Extensive exposure to configuration management policies along with automation of scripting using Bash/Shell scripting.
- Used Terraform for building, changing, versioning infrastructure and collaborating the automation of Azure Infrastructure via Terraform.
- Used Splunk to monitor the system logs as well as notify the incident management system upon exceeding thresholds.
- Coordinate/assist developers with establishing and applying appropriate branching, labeling/naming conventions using GIT source control and analyzing and resolve conflicts related to merging of source code for GIT.
- Maintained JIRA for tracking and updating project defects and tasks ensuring the successful completion of tasks in sprint.

Environment: Azure, Kubernetes, Docker, Terraform, GIT, Ansible, Maven, Nginx, Apache Tomcat, Python, Power Shell, Windows/Linux.

Jr. Linux Administrator Aug 2013 – Jan 2015

Astra Technology, Kathmandu Nepal

Responsibilities:

- Managed and maintained Linux-based servers, including Red Hat Linux environments, to ensure high availability and performance.
- Configuring the services for sharing the resources from Unix such as remote login through TELNET, FTP, SSH, SFTP, SCP.
- Developed and executed Bash and PowerShell scripts for routine tasks and system monitoring to streamline administrative tasks.
- Configured and managed version control using Git and GitHub, promoting collaboration and efficient code versioning within the team.
- Monitored and optimized network performance by troubleshooting TCP/IP connectivity issues and configuring network settings.
- Conducted basic MySQL database management tasks, including backups and queries, to support application requirements.
- Managing systems routine backup, scheduling jobs like disabling and enabling cron jobs, enabling system logging, network logging of servers for maintenance, performance tuning, testing.
- Hands on experience in automating builds and deployment process using Shell scripting.
- Utilized VMware to create, configure, and manage virtual machines, supporting server environments and optimizing resource utilization.
- Executing Kickstart configuration for automating Linux installations.
- Troubleshooting Linux network, security related issues, capturing packets using tools such as Iptables, firewall, TCP wrappers, NMAP.
- Employed Visual Studio for basic scripting and integration tasks, enhancing development and deployment processes

Environment: Kickstart, Linux, Tomcat, Jira, Confluence, VM Ware, Oracle, MySQL, Red hat.